

What is the aim of EcoTubeHQ?

YouTube users view a whopping 1 billion hours of video each day. That accounts for around 5 billion YouTube videos being watched each day. [\[comparitech\]](#)

1 billion hours of video streamed each day at 1080p requires a *lot* more data than video streamed at lower resolutions.

The table below gives the details - the data figures are derived from [eskimo.travel](#) and are approximate due to the multiple codecs used by YouTube.



['The True Scale of YouTube' - Click for video](#)

One exabyte (EB) is equal to one billion gigabytes (GB)

Resolution	Data per Hour	Total Data (1B hours)	% of 1080p
144p	0.1 GB	0.1 EB	3%
240p	0.2 GB	0.2 EB	6%
360p	0.4 GB	0.4 EB	12%
480p	0.6 GB	0.6 EB	18%
720p	2.0 GB	2.0 EB	61%
1080p	3.3 GB	3.3 EB	100%

EcoTubeHQ does not offer 1080p video as an output resolution, it only offers resolutions up to 720p...

Streaming one hour of 1080p YouTube video to a PC consumes approximately 0.15 to 0.35 kWh of electricity, broken down into roughly 0.03–0.06 kWh for network transmission and 0.10–0.25 kWh for powering your computer and monitor. This total is equivalent to boiling an electric kettle four to five times or running a standard LED bulb for 15–20 hours, with the device itself accounting for the majority of the energy use while the data traveling across the internet contributes a smaller, though measurable, portion.

...the aim of EcoTubeHQ is to minimize data traveling across the internet.

To make up for the reduced resolution the player utilizes multiple upscalers to enhance the video output in **Quality** mode. One of the upscalers used is ESPCN:

ESPCN_x2_8.gls is a fast, lightweight neural network upscaler designed to double the resolution of video (2x) by performing all complex calculations on the small, low-resolution image before efficiently shuffling the pixels to create a larger, sharper output.

The other upscalers used are variants of RAVU:

RAVU (Rapid and Accurate Video Upscaling) is a neural network-based upscaler that intelligently reconstructs missing details in low-resolution video by learning complex patterns from high-quality training data, rather than just guessing pixel values like traditional methods. It produces significantly sharper lines, smoother gradients, and fewer artifacts (like blurring or jagged edges) while maintaining a fast speed, making it ideal for transforming blurry SD content into crisp HD with a natural, high-fidelity look.

The end result of high quality upscaling is the quality of a 720p video is very close to that of a 1080p browser video.



Figure 1: Comparison of 1080p in a browser [left] with EcoTube 720p [right]

When video data is being downloaded hysteresis is utilised to prevent the constant download of data. A simple description of hysteresis being used in this manner is a 'buffer gap'.

A 'buffer gap' optimizes playback efficiency by avoiding constant data downloads; instead, the player initially fetches a 22-second chunk of audio and video data to create a buffer, it then pauses downloading entirely while playing back from this buffer until the data drops to 15 seconds, at which point it performs a quick bulk transfer to refill the buffer with another 7 seconds of data. This cycle of playback repeats throughout the video, allowing the CPU and network radio to "sleep" between transfers, which reduces power consumption and extends the battery life of laptops.

Another benefit of this download method is it only uses 22 seconds of data for a buffer, and this is considerably less than the standard 90 second YouTube buffer, meaning if a video is stopped before completion then considerably less data is wasted.

To further minimize data downloaded the player allows av1 to be output, if av1 isn't available then vp9 is output, and if vp9 isn't available then the player falls back to h.264.

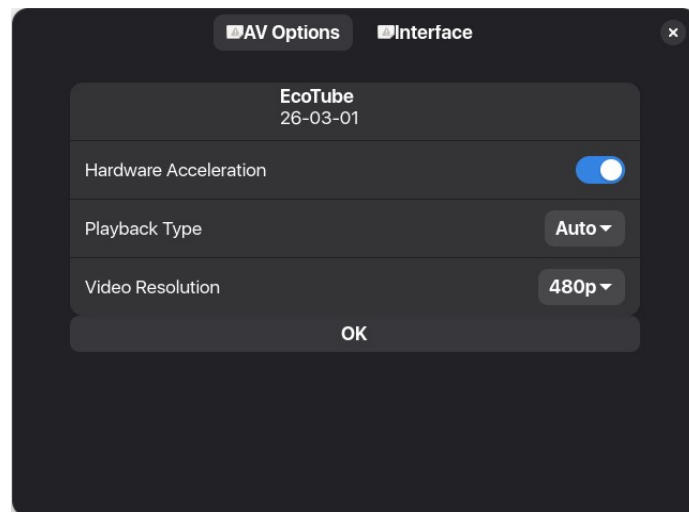
av1, vp9, and h.264 are video compression standards that determine how efficiently video files are shrunk for streaming or storage, with h.264 being the oldest and most universally compatible (working on almost every device but less efficient), vp9 being a modern successor developed by Google that offers better quality at lower file sizes (common on YouTube) but requires newer hardware to decode smoothly, and av1 being the newest, most advanced open-source codec that delivers the highest quality with the smallest file sizes but requires more computing power to play videos.

The following table compares the codecs.

Video Codec Comparison			
Aspect	AV1	VP9	H.264
Compression Efficiency	Best — ~30% better than VP9, ~50% better than H.264 at same quality	Good — ~25% better than H.264	Baseline — less efficient, requires higher bitrates for equivalent quality
Decoding Speed	Moderate — reasonably fast on modern hardware; hardware decoders increasingly common	Moderate to Good — hardware support widespread	Excellent — hardware support nearly universal; extremely fast

Player Preferences

Click on the hamburger menu next to the minimize button, then select Preferences. This shows the following...



...here's what it means:

Hardware Acceleration

On [Default] - Vulkan GPU Compute Acceleration offloads decoding to your GPU for av1, vp9, h.264, and h.265 video codecs, reducing CPU usage. This improves playback on older PCs and extends laptop battery life. Disable if you experience playback problems.

Playback Type

Auto [Default] - **Quality** is the default setting on desktop computers. Laptops use **Quality** when plugged in and automatically switch to **Powersave** on battery power.

In the **Quality** mode EcoTubeHQ focuses on video quality whilst minimizing data requirements:

- Maximum resolution 720p
- High quality upscaling
- av1 is used by default
- If av1 is not available then vp9 is used
- If vp9 is not available then h.264 is used

In the **Powersave** mode EcoTubeHQ focuses on power saving:

- Maximum resolution 480p
- Fast upscaling
- vp9 is used by default
- If vp9 is not available then h.264 is used

EcoTubeHQ further reduces data usage by not outputting 720p video at 60fps as this requires an additional 50%+ of data when compared to 30fps.

Note: YouTube restricts some old h.264 videos to three resolutions: 144p, 360p, and 720p. If EcoTubeHQ is set to output an unavailable h.264 resolution [240p or 480p], it automatically switches to the next highest tier - 240p outputs video at 360p, and 480p outputs video at 720p.

The end result is optimal video quality with minimal data usage.

Video Resolution

Supported Video Resolutions: 720p, **480p [Default]**, 360p, 240p, 144p

The **Quality** playback mode restricts the maximum video resolution to 720p to minimize data usage, and **Powersave** caps the resolution at 480p to minimize CPU usage.

Note: EcoTubeHQ automatically updates yt-dlp and FFmpeg to minimize issues.

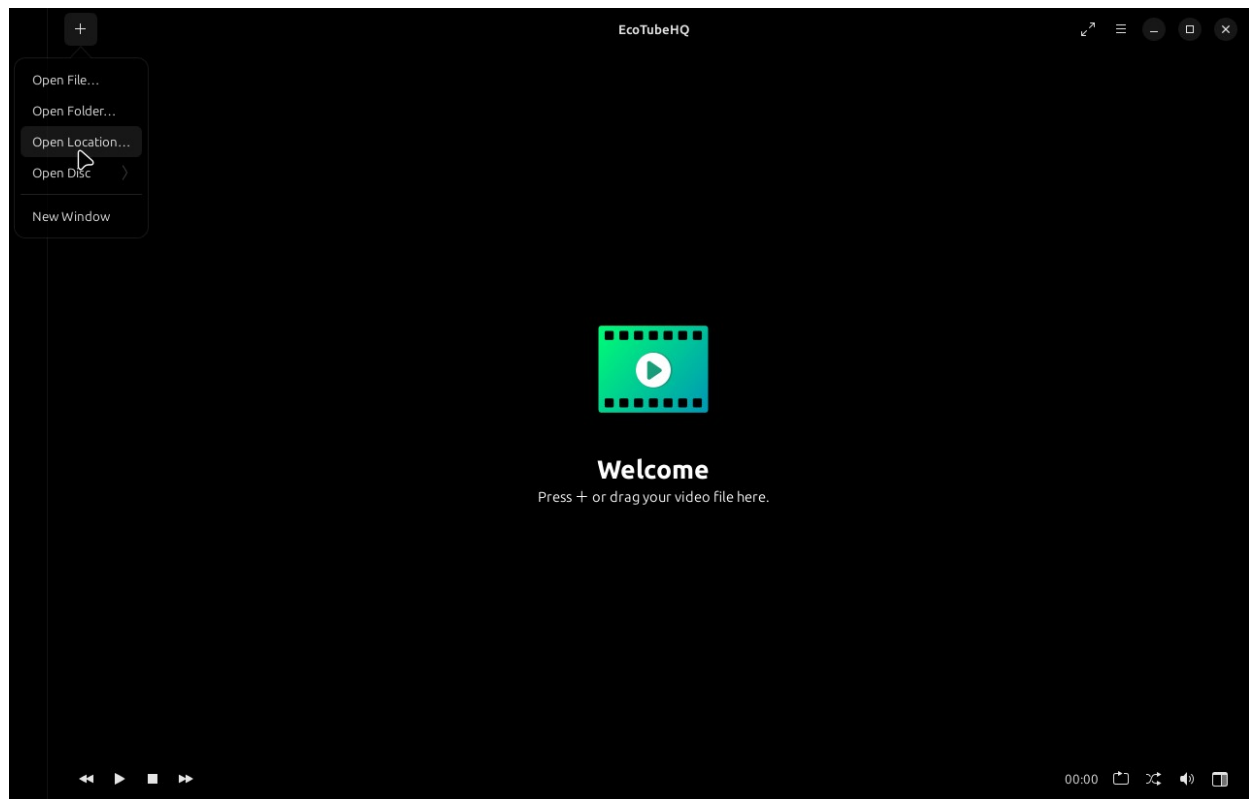
*If the player regularly displays the error '**Playback was terminated abnormally. Reason: loading failed**' when you try and play a video then there may be a problem with the streaming server or the software may need to update. Simply close the player for a while and try again later.*

Note 2: Using EcoTubeHQ via a VPN can cause playback issues:

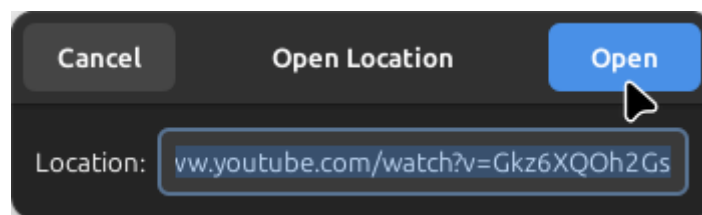
A VPN can cause YouTube playback interruptions primarily because it introduces additional network overhead: your traffic travels through an encrypted tunnel to a VPN server before reaching YouTube, which increases latency and potential points of failure. This extra routing can reduce available bandwidth, especially if the VPN server is geographically distant or oversubscribed, causing buffering when YouTube's adaptive bitrate streaming detects slower connections. Additionally, YouTube may actively throttle or block traffic from known VPN IP addresses to enforce regional restrictions or prevent abuse, leading to connection timeouts or degraded stream quality. Finally, packet loss within the VPN tunnel itself - due to congestion or poor server infrastructure - can trigger retransmissions that further disrupt the smooth data flow required for uninterrupted video playback.

How to use

To play a video from YouTube / Twitter X simply copy the URL from the browser, click '+' at the top left of the player and select 'Open Location'...



...the player automatically places the copied video location into the 'Location' entry box...



...click the 'Open' button to start playback. This will also initiate the playback of livestreams which will generally be output using the h.264 codec.

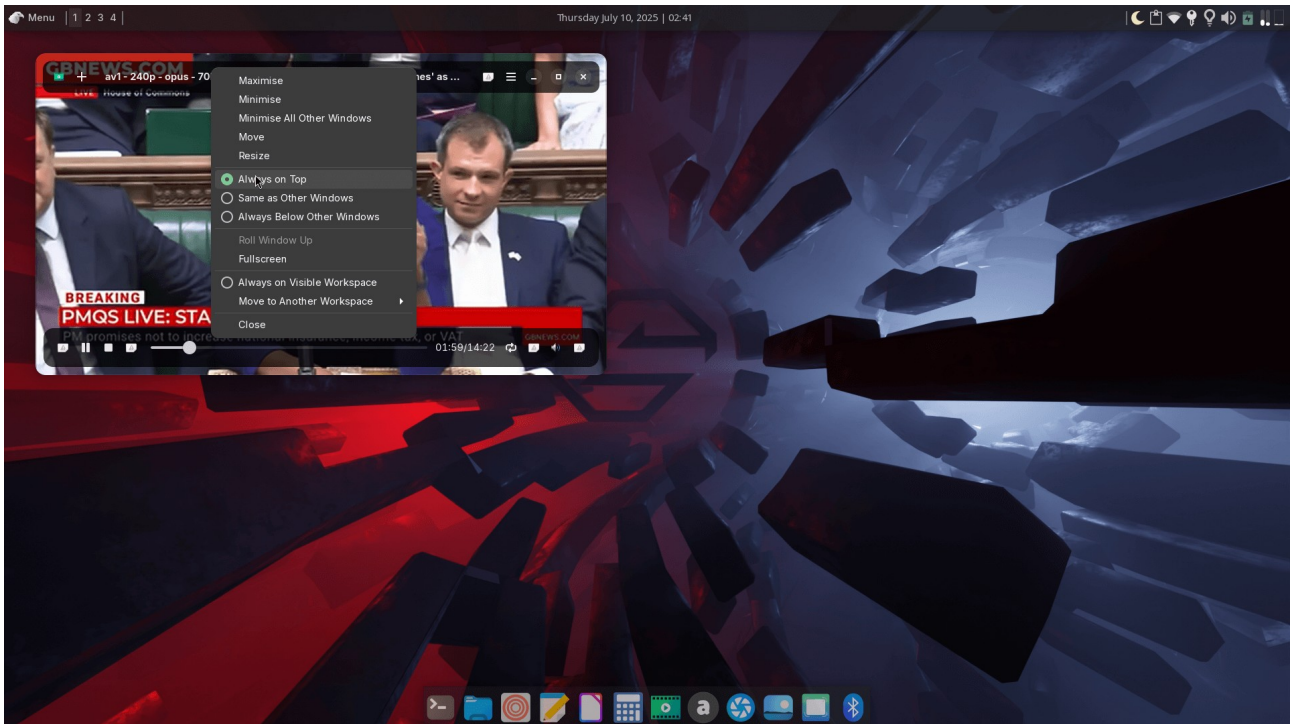
- Press the space bar or right click the mouse on the player screen to pause / play video playback.
- Rotate the mouse wheel on the playback window to turn the audio volume up / down – the volume will reset to normal when the player is restarted.
- Double click the player window during playback to play the video fullscreen.
- Press 'Esc' or double click on the player window during fullscreen playback to return the player to non-fullscreen playback.

Miniplayer Usage

If you are watching a YouTube video which does not require fullscreen or you want to minimize CPU usage during playback then set the player's video output resolution to 144p or 240p.

This automatically reduces the size of the player's non-fullscreen window, effectively providing you with a miniplayer. The size of 240p videos is shown in the screenshot below, and 144p videos are displayed in a smaller window.

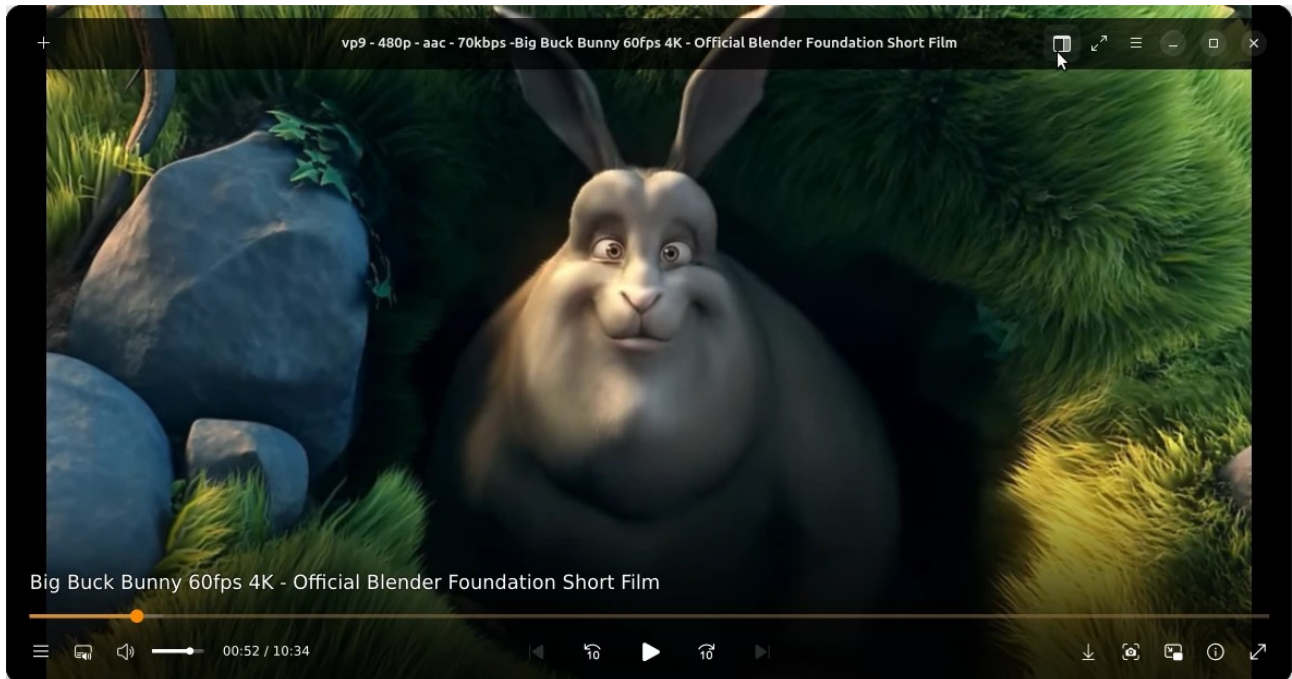
To keep the player on top of all other windows during playback right-click on the top bar and select 'Always on top' as shown below.



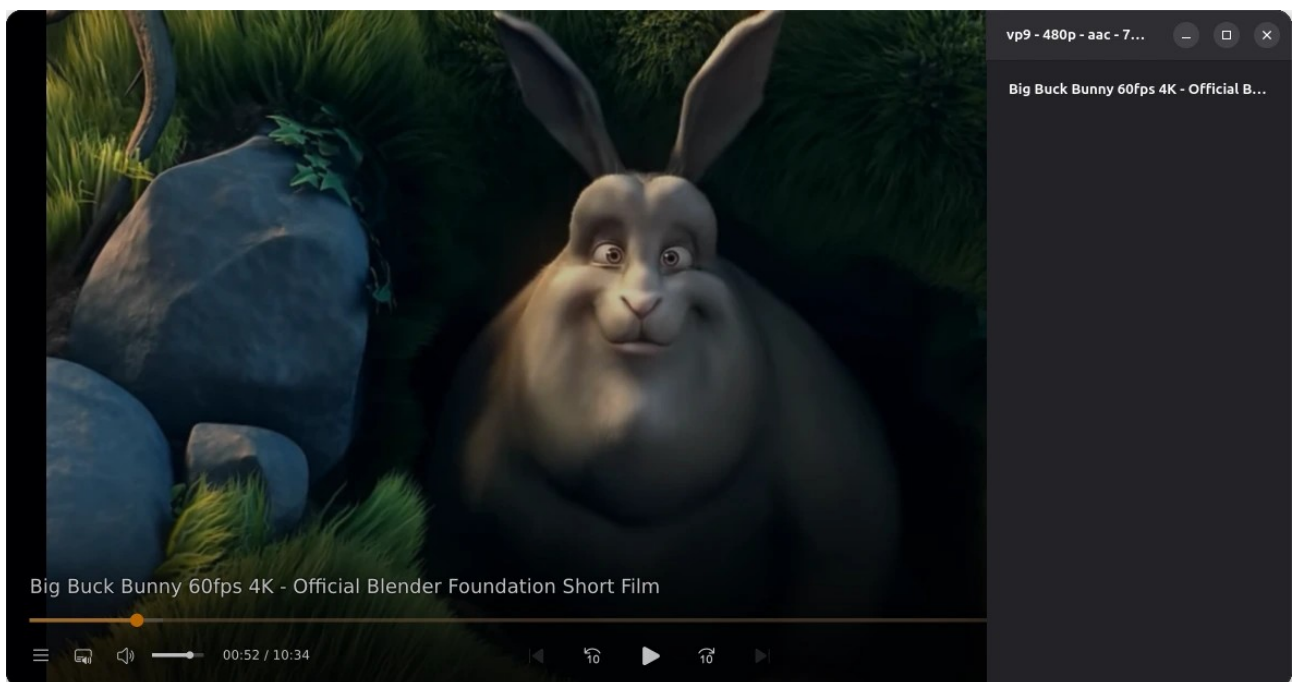
The miniplayer significantly reduces the amount of data used for videos of things such as podcasts where video quality is not paramount.

* To further reduce CPU usage set the Playback Type to **Powersave**

Playlists



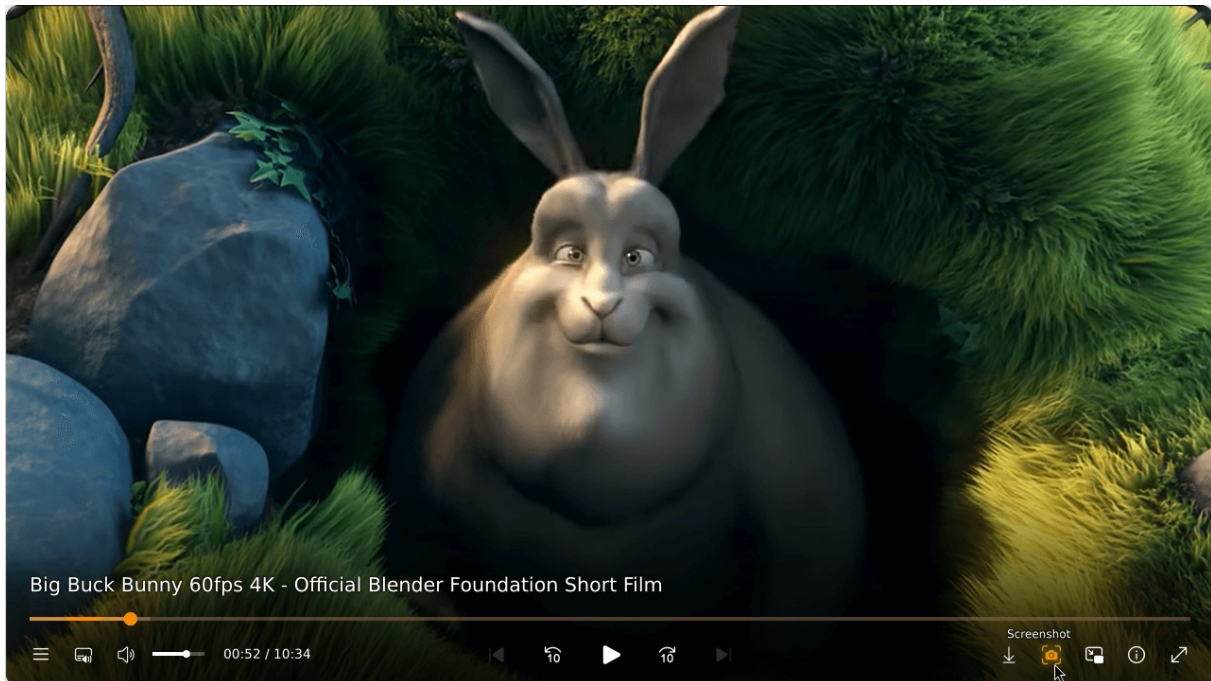
To add a video to a playlist click on the top icon indicated above. When this is done the playlist opens.



To add videos from an external streaming service drag and drop the browser URL for the playlist onto the player's playlist. Additional videos can be added prior to, and during playback. To start playback simply double-click on the required playlist video.

Screenshots

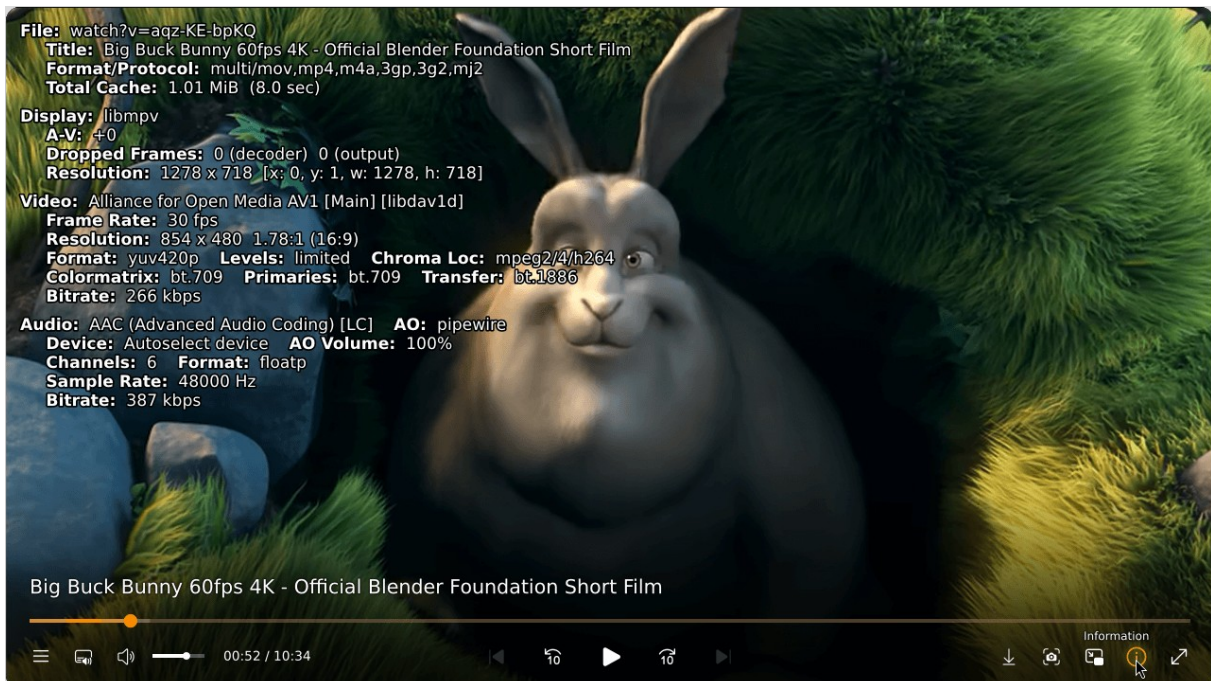
To take a screenshot of the on-screen picture click the button at the bottom of the player shown below...



...the image will be stored in the **Pictures** folder.

Audio and video information

To see information relating to the on-screen video click the information button shown below...



Additional shaders used by EcoTubeHQ

dehaasn_ads_luma_RT.glsl is a lightweight, real-time shader that removes visible "banding" (color stripes in smooth gradients like skies) and compression artifacts from video by intelligently smoothing only the brightness (luma) channel. Unlike a simple blur that softens the entire image, it uses an adaptive algorithm to detect flat areas where banding occurs and gently blends them out while preserving sharp edges and fine details, resulting in a cleaner, smoother picture without losing clarity or wasting GPU power on the color channels.

colorlevel_expand_luma.glsl is a lightweight shader that recovers lost detail in dark shadows and bright highlights by stretching the video's brightness levels to fill the full range of your display. Unlike a simple brightness slider that washes out the whole image, it specifically targets "crushed" blacks and "blown" whites to reveal hidden textures and depth in those areas while leaving the mid-tones and colors untouched, resulting in a more natural, high-contrast picture with better visibility in both dark and light scenes.

eq_clarity_RT.glsl is a real-time shader that enhances the perceived depth and texture of an image by boosting local contrast in the mid-tones without creating the harsh halos or "ringing" artifacts common with standard sharpening filters. Instead of just making edges crisper, it makes details like skin pores, fabric weaves, and foliage look more three-dimensional and natural, giving the video a richer, more "punchy" appearance while preserving the smoothness of flat areas and avoiding the artificial, over-processed look of aggressive sharpeners.

kVibrance_RT.glsl is a smart color-enhancement shader that boosts the saturation of dull, muted colors to make the image look more lively and vibrant, while intentionally leaving already-rich colors and skin tones untouched to prevent them from becoming oversaturated or unnatural. Unlike a standard saturation slider that intensifies every color equally (often turning skin orange or skies neon), this tool selectively targets only the grayish or washed-out areas to restore their richness, resulting in a more balanced and visually appealing picture that feels fresh without looking artificial.

PixelClipper_RT.glsl is a real-time cleanup shader that removes unwanted "halos" or glowing fringes (ringing artifacts) that often appear around sharp edges after upscaling or sharpening. It works by detecting pixels that have incorrectly "overshot" their natural brightness or color values and gently pulling them back to the correct range, effectively erasing the ghostly outlines without blurring the actual image details, resulting in a clean, crisp picture free from distracting visual noise.

noise_static_luma.glsl is a shader that adds a subtle layer of fine, static-like grain to the brightness channel of the video to mask visible "banding" (color stripes) and restore natural texture. By injecting a tiny amount of random noise, it breaks up smooth gradients that look artificial or plastic after upscaling or heavy denoising, making the image appear more organic and film-like without introducing colorful speckles or significantly impacting performance.

EcoTubeHQ concept and configuration: Colin Bett

Coding and additional ideas: Sako Adams

Forked from the Celluloid Video Player

This application comes with absolutely no warranty. See the GNU General Public Licence, version 3 or later for details - <https://www.gnu.org/licenses/gpl-3.0.html>.

Changelog

26.09.02

- Fixed preconfiguration data

26.09.01

- Externalized the mpv configuration
- Auto customize player with better video quality and hardware usage
- Added new option in preferences box
- Improved caching and buffering

26.07.02

- Preview thumbnail disabled on streaming sources
- Added audio only option
- Eliminated stop/slow buffering on playback pause
- Added option to enable / disable video grain

26.07.01

- Stops downloading when the cache is full
- Remove static bitrate
- Updated player info

26.06.02

- Added internal pdf viewer
- Allow 720p in auto mode when laptop is plugged in
- Fix FSR when laptop is plugged in
- Fix resolution fallback when plugged in

26.06.01

- Fix format selection, take into account when audio is not in separate file
- Control end of playback
- Fix playback per block of 10 sec
- Improve UI

26.04.01

- Updated Gnome runtime to the latest version
- Fixed format selection
- Improved video loading

26.03.01

- Updated Preferences and removed HDR option
- Changed video upscale option to nearest neighbor
- Fixed format selection
- Changed app name in Start Menu / launcher

v26.02.01

- Updated preferences and removed deprecated features
- Fixed dark theme issue
- Fixed issue occurring on Wayland on Open Location option
- Auto update ffmpeg

v25.12.02

- Add option to set many seconds of audio/video to prefetch
- Added mpv reload script
- Update ecotube fidelityfx super resolution logo

v25.12.01

- Fixed thumbnail on sandbox
- Disabled video download
- Enhance interface
- Reorganized Auto mode

v25.11.01

- Enhance interface
- Added modern OSC with screenshot and stats
- Fix FSR when plugged in
- Add 'auto' mode in desktop instance

v25.10.2

- yt-dlp is auto-updated when a new release is available
- Added modern On Screen Control
- Local files show thumbnails on the seekbar
- Download the video
- Enhance UI

v25.10.1

- Added 'auto mode' in playback type
- Fix stream latency
- Enhance UI

v25.9.3

- Updated yt-dlp
- Bilinear upscaling is used for powersave mode
- Enhance UI

v25.9.2

- Fix codec selection in powersave mode
- Updated playlist mode
- Enhance UI

v25.9.1

- Updated playback options - Powersave and Quality
- Updated playlist mode
- Added HDR option in preference
- Enhance UI
- Fixed memory usage in startup

v25.8.2

- Added two playback options - Powersave and Quality
- Removed redundant codec selection
- Reorganized default playlist
- Included no https protocol in format selection
- Reduced network stream usage

v25.8.1

- Forced yt-dlp to use MPEG-DASH on some resolutions
- Reduce stream when paused
- Fixed resolution dropping
- Exclude HDR
- Add playback type

v25.7.2

- Forced mpv to avoid HLS - temporarily

v25.7.1

- Added Vulkan support
- Fixed video / audio download + improved video loading
- Fix format selection
- Updated caching system
- Fixed auto-resize player

v25.5.1

- Upgrade to gtk4.16 and libadwaita-1 1.6
- Enhance interface with playlist overlay
- New update notification
- New dialog for "How to use it"
- Minor playback fix

v25.3.3

- Buffering: prefetch is now limited to 10 seconds to minimize data used when videos are not played in their entirety.
- Hardware acceleration is automatically activated if a laptop is running on battery power.
- Auto update parsers.
- Updated the way default resolutions are selected.
- The preference interface has been updated.

V25.2.1

26 February 2025

Initial conversion from Celluloid.

Thanks to [haggen88](#) for the icon update and [albanobattistella](#) for the patches,

14 August 2024
Initial commit to Github.